Sequences — problem 9.49

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Of the sequence $a_0, a_1, \ldots, a_n$ it is known that $a_0 = a_n = 0$ and that $a_{k-1} - 2a_k + a_{k+1} \geq 0$ for all $k = 1, \ldots, n-1$. Prove that $a_k \geq 0$ for all k.